A project report on

AFFORDABLE AUTOMATED INDOOR PLANT MONITORING SYSTEM FOR VISUALLY IMPAIRED

Submitted in partial fulfillment for the award of the degree of

Bachelor of Technology in Computer Science and Engineering

by

ADITI SINGH (21BCE1376)

ADITTYA NARAYAN (21BCE1672)

ANJANA SATHISH AARTHY (21BCE5632)

SCHOOL OF COMPUTER SCIENCE AND ENGINEERING

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Vellore Institute of Technology
(Deemed to be University under section 3 of UGC Act, 1956)
CHENNAI

DECLARATION

I hereby declare that the thesis entitled "AFFORDABLE AUTOMATED INDOOR PLANT MONITORING SYSTEM FOR VISUALLY IMPAIRED" submitted by Adittya Narayan (21BCE1672), for the award of the degree of Bachelor of Technology in Computer Science and Engineering, Vellore Institute of Technology, Chennai is a record of Bonafide work carried out by me under the supervision of Dr. Umamaheswari E.

I further declare that the work reported in this thesis has not been submitted and will not be submitted, either in part or in full, for the award of any other degree or diploma in this institute or any other institute or university.

Place: Chennai

Date: 14-Apr-25

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Signature of the Candidate

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CERTIFICATE

This is to certify that the report entitled **"Affordable Automated Indoor Plant Monitoring System For Visually Impaired"** is prepared and submitted by **Adittya Narayan (21BCE1672)** to Vellore Institute of Technology, Chennai, in partial fulfillment of the requirement for the award of the degree of **Bachelor of Technology in COMPUTER SCIENCE AND ENGINEERING** is a bonafide record carried out under my guidance. The project fulfills the requirements as per the regulations of this University and in my opinion meets the necessary standards for submission. The contents of this report have not been submitted and will not be submitted either in part or in full, for the award of any other degree or diploma and the same is certified.

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ABSTRACT

This system presents an IoT-powered indoor gardening solution specifically designed to enable visually impaired individuals to cultivate and maintain plants independently. By integrating automation, machine learning, and a user-friendly mobile application, the system simplifies plant care through real-time feedback and intelligent insights. The system continuously monitors plant health and growth using image-based analysis, while tracking essential environmental factors such as light levels, temperature, humidity, and soil moisture. Users receive personalized recommendations through voice assistance and tactile vibrations, ensuring seamless interaction without the need for visual interfaces.

A key innovation of this system is the inclusion of a security feature utilizing infrared (IR) sensors, which detects nearby movement and sends alerts to the mobile application, enhancing personal safety for visually impaired users. The system prioritizes affordability, accessibility, and ease of use, making it particularly effective for growing indoor plants such as herbs and leafy greens. Beyond personal gardening, this solution extends to therapeutic gardening, educational programs, and urban indoor farming, offering a sustainable and inclusive approach to indoor plant care. By leveraging AI-driven automation and IoT connectivity, this system empowers users with greater independence and a more enriching gardening experience.

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LIST OF ACRONYMS

YOLO –	You Only Look Once
ViT –	Vision Transformers
ESP –	Espressif Systems Product
IR –	Infrared
LDR –	Light Dependent Resistor
API –	Application Programming Interface
LLM –	Large Language Model
DHT –	Digital Humidity and Temperature

Chapter 1
Introduction

1.1 INTRODUCTION

Indoor gardening has become increasingly popular because it can offer fresh fruits and vegetables, purify the air, and promote mental health. For the visually impaired, though, having indoor plants is not always easy because one cannot see plant conditions and the environment. The Affordable Automated Indoor Plant Monitoring System for Visually Impaired will meet these challenges by combining smart IoT-based solutions to sense plant conditions, automate necessary care procedures, and offer an inclusive user interface with voice commands and haptic feedback. The system uses multiple sensors to determine soil moisture, temperature, humidity, and light intensity.

Moreover, an ESP32-CAM module is utilized for image-based monitoring of plant health, using machine learning to identify diseases and fruit presence. Automated watering systems provide plants with proper hydration without the need for user intervention. The system also features a security aspect in the form of an infrared (IR) sensor on the pot to identify possible intrusions, adding an extra layer of protection for users. The information gathered by these sensors is pushed to a mobile application meant for visually impaired users. The application gives live updates in the form of voice cues, and upcoming developments will include vibration alerts and a voice chatbot for voice interaction. Additionally, an audio SOS function is in the pipeline, which will identify emergency calls and send out emergency alerts. By providing easier plant monitoring, the project encourages inclusivity and the independence of the visually impaired to maintain indoor gardens.

1.2 OVERVIEW OF AUTOMATED PLANT MONITORING SYSTEMS

Computerized plant monitoring systems leverage IoT and AI technologies to automate plant care through constant monitoring of environmental parameters and taking requisite measures. These systems may encompass:

- **Environmental Monitoring Sensors**: Measuring such parameters as soil moisture, temperature, humidity, and light intensity to ascertain plant requirements.

- **Automatic Watering Systems**: Utilizing pumps and motors to deliver water as per soil moisture conditions.

- **Machine Learning for Plant Health Monitoring**: Disease detection, growth patterns, and fruiting status from analyzing plant images.

- **Mobile App Integration**: Real-time data display, alerting, and remote operation of plant care systems.

While there are a number of automated gardening solutions, most do not have accessibility features for visually impaired gardeners. Our system fills this particular gap by incorporating voice aid, security systems, and possible vibration feedback.

1.3 CHALLENGES IN INDOOR GARDENING FOR VISUALLY IMPAIRED INDIVIDUALS

Indoor plant maintenance poses several issues for the visually impaired, mainly based on their use of non-visual information in caring for plants. The most notable is a lack of ability to visually examine the condition of plants. Symptoms of illness, water deficit, or mineral deficiencies—like wilting of the leaves, color changes, or fungal attacks—are usually identified by vision. Without visual inspection, visually impaired people cannot easily ascertain if a plant needs attention.

Another significant challenge is uncertainty when it comes to watering schedules. Because overwatering and underwatering can be equally harmful to plant health, gardeners generally use visual cues like soil dryness or drooping leaves to determine watering frequency. Without readily available alternatives, visually impaired people have to depend on memory or external help, which can cause irregular plant care.

Furthermore, current smart gardening solutions are largely visual in nature, with visual interfaces like mobile applications or LED lights, thus being inaccessible to visually impaired individuals. Even when these systems alert, they are usually in the form of text messages and not voice or haptic feedback. The absence of real-time, accessible notifications also makes plant care more difficult, as users will not be timely informed about environmental conditions or interventions required.

In order to overcome these challenges, an automated and accessible indoor gardening system is required. By incorporating voice guidance, real-time monitoring of data, and automated plant care systems, a system can enable visually impaired users to take care of their plants on their own and effectively.

1.4 PROBLEM STATEMENT

Existing indoor gardening systems do not integrate accessibility features, which makes it difficult for visually impaired persons to check and ensure plant wellness by themselves. Conventional methods tend to depend on visual indicators, which present a limitation for visually impaired individuals. The demand is for a cost-effective IoT-based system that can automatically perform necessary plant care functions like watering, fertilization, and disease monitoring. Moreover, the system must offer instant, accessible feedback in the form of voice aid, vibrations, and other visual-independent alerts. Through the inclusion of smart sensors, automation, and AI-powered plant health tracking, this solution seeks to enhance indoor gardening accessibility and ease for visually impaired individuals.

1.5 OBJECTIVES OF THE PROJECT

This project seeks to create an Affordable Automated Indoor Plant Monitoring System for Visually Impaired with the following goals:

- Develop an IoT-enabled system to continuously monitor key environmental parameters such as soil moisture, temperature, humidity, and light intensity.
- Design an automated irrigation mechanism that operates based on real-time input from soil moisture sensors.
- Integrate an ESP32-CAM module to periodically capture images of the plant for machine learning-based analysis of disease and growth patterns.
- Build an accessible mobile application equipped with voice assistance to provide real-time updates and insights on plant health.
- Incorporate an IR-based intruder detection system to enhance security and notify users of unauthorized movement near the plant setup.
- Explore additional accessibility enhancements, such as vibration feedback and a voice-interactive chatbot to offer plant care guidance.
- Evaluate the feasibility of an audio-based SOS system capable of detecting distress sounds and triggering emergency alerts to designated contacts.

1.6 SCOPE OF THE PROJECT

This project seeks to create a low-cost, accessible, and easy-to-use indoor plant monitoring system tailored for visually impaired users. The system combines different hardware and software elements to make plant care automatic while providing real-time, accessible feedback.

The hardware development feature comprises sensor integration for tracking environmental parameters, an ESP32-based system for data acquisition, and an automated watering system to keep the soil at optimal moisture levels. These features combine to minimize the need for human intervention and provide plants with proper care.

Software-wise, the system has a smartphone app with voice support and future haptic output, allowing easier user interaction through alerts and facilitation of communications with the system. Machine learning is used primarily in plant disease and fruit identification models, wherein image-based classification models analyze pictures taken to gather information on the condition of plants.

Security is also a key feature, where an intruder detection system based on an IR sensor is provided to maximize personal safety. Furthermore, accessibility improvements like voice commands, chatbot interfaces, and emergency alert systems maximize the usability of the system for visually impaired persons.

Although this project presents a viable indoor gardening system, it is not an agricultural automation solution for large-scale farming. Rather, it is an indoor garden on a small scale for individual use, providing a cost-effective and convenient method of enabling independent plant care.

1.7 TARGET USERS

This project is mainly intended for people with visual impairments who want to do independent indoor gardening. Automating necessary plant care operations and giving real-time information in the form of voice assistance as well as other accessible alerts, the system makes it possible for them to take care of plant health without visual assistance.

Aside from visually challenged users, the system also addresses the needs of elderly people who might have trouble tending and monitoring plants by hand. Automation of watering, environmental monitoring, and easy notification by the project minimizes maintenance effort for the plants.

Plant lovers seeking an inexpensive, automated system with real-time monitoring and alerts can also benefit from this project. IoT and machine learning integration ensures plant health is always monitored, providing users with insights and timely interventions.

Finally, the project is a useful resource for developers and researchers who are interested in IoT-based assistive technology for smart home use. By catering to the particular needs of visually impaired users, the system encourages inclusivity and increases accessibility in indoor gardening, providing a basis for further innovation in assistive smart home solutions.

Chapter 2

Background

2.1 INTRODUCTION

This chapter provides an overview of background concepts necessary to understand the development and implementation of the Affordable Automated Indoor Plant Monitoring System for Visually Impaired. It covers the significance of indoor gardening, the challenges faced by visually impaired individuals in maintaining indoor plants, and the potential of automation and IoT technologies to address these challenges.

2.2 IMPORTANCE OF INDOOR PLANT MONITORING

Indoor plants are found to purify the air, decrease stress levels, and promote mental health. They have an important role in cities where there can be a lack of exposure to nature. By giving a natural look, indoor plants can help create a peaceful environment that helps with emotional well-being. Indoor gardening must be done carefully while keeping environmental factors like soil water, temperature, humidity, and light under observation. These elements are critical to healthy plant development and can have a substantial impact on the health of both the plants and their owners. In addition, indoor gardening can encourage a feeling of accomplishment and responsibility, as well as a sense of connection to nature, particularly for visually impaired individuals who might otherwise feel isolated from the natural environment.

2.3 CHALLENGES FACED BY VISUALLY IMPAIRED INDIVIDUALS

Visually impaired individuals encounter several challenges when attempting to care for indoor plants, including:

- **Difficulty in Assessing Plant Health:** Without visual cues, individuals cannot easily determine the health of their plants, identify signs of disease, or assess their growth conditions. This limitation can lead to neglect or improper care, resulting in plant deterioration.

- **Inefficient Watering Practices:** Traditional watering schedules may lead to overwatering or underwatering, as they rely on memory or assistance from others. This can affect plant health, as both conditions can cause root rot or dehydration, respectively.

- **Inaccessibility of Existing Technologies:** Many current indoor gardening systems lack accessibility features, making them unsuitable for visually impaired users. These technologies often present data in visual formats, excluding those who cannot see.

- **Limited Awareness of Plant Needs:** Without real-time monitoring, it is challenging to understand when plants require water, nutrients, or other care. Users may miss critical indicators of plant stress, leading to further complications in plant health management.

2.4 POTENTIAL OF AUTOMATION AND IOT IN PLANT CARE

The integration of automation and IoT (Internet of Things) technologies can significantly enhance indoor gardening experiences for visually impaired individuals. Key benefits include:

- **Real-Time Monitoring:** IoT sensors can continuously track soil moisture, temperature, and humidity, providing users with instant updates about plant conditions.

- **Automated Irrigation:** Smart systems can water plants automatically based on real-time data, reducing the burden of manual care.

- **Remote Access and Control:** Users can access plant health information and control systems remotely through mobile applications, making gardening more convenient and engaging.

- **Voice Assistance:** Integrating voice technology can provide users with auditory feedback about their plants, ensuring that visually impaired individuals can interact with their gardening systems effectively.

2.5 MACHINE LEARNING FOR PLANT HEALTH MONITORING

Machine learning technologies can enhance plant monitoring systems by enabling image analysis and disease detection. Using the YOLOv11 model, an ESP32-CAM module captures images of plants, which are processed to identify signs of diseases or pests. The model's ability to detect objects and classify images allows it to discern healthy plants from those that may be suffering from various conditions. This approach leverages existing machine learning models to provide actionable insights, helping users take timely action to maintain plant health. By utilizing pre-trained models, the system minimizes the computational load and accelerates the detection process, allowing users to respond quickly to emerging issues with their plants.

2.6 SECURITY CONSIDERATIONS

In addition to monitoring plant health, safety features are essential for visually impaired users. Implementing an infrared (IR) sensor can serve as an intruder detection mechanism, alerting users to unexpected movements around their plants. The strategic placement of the IR sensor in the pot ensures that the feature remains discreet, as it would not draw attention from potential intruders. Additionally, an audio SOS feature can be integrated to detect distress sounds, such as cries for help, and send alerts to emergency contacts. This dual security approach aims to provide peace of mind for users, allowing them to engage in indoor gardening without concerns about personal safety.

Chapter 3
Literature Survey

3.1 SMART FARM CONTROL SYSTEM

This patent is for an IoT-based system to automate plant growth in a sealed agricultural environment. The system combines facility control equipment like shading devices, irrigation systems, ventilation units, temperature and humidity controllers, oxygen concentration controllers, and nutrient supply systems. A central IoT controller processes real-time environmental data and dynamically changes operational schedules depending on temperature fluctuations. By sensing the variations in external and internal temperatures, and deviations from predetermined optimal conditions, the system regulates ventilation, shading, and temperature settings to achieve optimal growing conditions for crops. The smart farm control system provides automated environmental control with minimal human intervention, thereby ensuring optimal crop growth.

It continuously monitors parameters like temperature, humidity, soil moisture, and gas concentrations by an intercommunicating network of sensors. Real-time sensing and decision-making enable the system to dynamically adjust device operations, boosting energy efficiency. Moreover, a database server enables remote management with automatic updating and synchronization of control schedules over multiple IoT devices. Deviation- based decision-making implementation guarantees that ecological adjustments are data- driven, avoiding uncertainties that can be detrimental to plant health and yield. In spite of its sophisticated automation, the patent does not have machine learning-based monitoring of plant health or predictive analytics. The system is based on rule-based automation instead of AI-based insights for plant disease diagnosis or growth prediction.

In addition, in spite of regulating environmental conditions effectively, it lacks features specifically designed for increased accessibility, including voice control or haptic feedback. Another issue is the system's reliance on sensors' accuracy and calibration; drift in sensor output over time could result in the wrong environmental corrections. Also, dependence on the availability of steady internet and electricity supply presents hindrances for round-the-clock operations. The patent is mainly concentrated on preconfigured control logic and does not much emphasize adaptability to different plants with dissimilar growth requirements. High initial installation expenses, such as sensors, control units, and networked devices, may also act as a deterrent to adoption, especially for small-scale farmers or individual consumers. In spite of these constraints, the system offers a very organized method of automated plant growth, providing useful insights into IoT-based environmental control in smart farming systems.

3.2 INDOOR HORTICULTURE CLIMATE CONTROL SYSTEM

The present patent describes an indoor climate control system to achieve maximum plant growth indoors through controlling temperature, humidity, and air quality. The system incorporates heating, cooling, and dehumidification into one unit with lower energy expense and increased environmental stability. Air-to-air heat exchangers, liquid heat exchangers, and cooling fluid circuits are used to ensure efficient airflow control and maintaining desired conditions. It uses different sensors to track temperature, humidity, and airflow rate, with the electronic controller dynamically regulating climate parameters. The automation supports several dehumidification modes, including cooling with or without dehumidification and heating with or without moisture control, enabling flexible climate adjustment. The system also recovers waste heat from artificial lighting for enhancing energy efficiency. Though the patent does not directly incorporate AI for monitoring plant health, its sensor-based automation can be supplemented with machine learning algorithms for predictive analysis. The efficiency of the system is that it keeps growing conditions stable and minimizes manual adjustments. Still, issues like high energy expenditure for dehumidification, dependency on liquid chillers, and complicated climate adaptation remain. Despite not targeting visually impaired accessibility, its automatic environmental controls could also indirectly help users by reducing direct plant maintenance. Having all heating, cooling, and dehumidification functions integrated into a single unit improves scalability for different indoor horticulture uses, so it is a good universal solution for controlled plant growth facilities.

3.3 AN ARDUINO-BASED AUTOMATED GARDENING SYSTEM FOR EFFICIENT AND SUSTAINABLE PLANT GROWTH

The research work entitled "An Arduino-Based Automated Gardening System for Efficient and Sustainable Plant Growth" introduces a novel automated gardening system consisting of an Arduino Nano, soil moisture sensor, relay module, and water pump. This system is aimed at maintaining soil moisture at optimal levels, thus ensuring a healthy growth of plants while saving water resources. Automated irrigation lessens the workload of manual intervention, leading to greener gardening practices.

The study discusses a number of technologies applicable to automated gardening, but with emphasis on how IoT can make gardening more efficient. It points out the application of IoT-based hydroponic systems and intelligent greenhouses that utilize microcontrollers, for example, Arduino and Raspberry Pi, in conjunction with an assortment of sensors used to monitor temperature, humidity, and light levels. These sensors interact with actuators to regulate vital functions such as irrigation and ventilation.

Even though the paper does not directly talk about assistive features for visually impaired users, it indicates that remote plant monitoring and automation may be modified to enhance accessibility. The use of soil moisture sensors by the system allows it to sense dryness and automatically irrigate, providing efficient water management through IoT-based mobile applications.

Although the paper is short on details about the possibility of applying machine learning and artificial intelligence, it does not specify models or applications for plant health monitoring. The results prove that the automated irrigation system can ensure proper moisture levels and prevent water wastage. The relay module controlled by Arduino effectively manages the water supply, providing constant hydration to the plants.

Though the positive points have been noted, the paper cites some challenges. The system greatly depends on the soil moisture sensor accuracy, which is prone to variations in environment and soil conditions. Moreover, the fixed threshold-based irrigation scheme might not prove to be adaptive for the wide-ranging requirements of various plant types. The lack of real-time cloud-based monitoring and control constrains the functioning of the system, and extension to larger indoor gardens might call for extensive reconfigurations.

In summary, although the automated gardening system demonstrates the advantages of employing Arduino and IoT technologies, there are still improvements to be made, especially in terms of assistive functions for visually impaired users and the incorporation of AI for more adaptive plant care.

3.4 SMART GARDENING SYSTEM USING IOT

The research paper "Smart Gardening System Using IoT" delves into the application of Internet of Things (IoT) technology in urban agriculture with an emphasis on maximizing sustainability, resource efficiency, and automation.

The system utilizes sensors and actuators to monitor environmental parameters in real- time, such as soil moisture, temperature, humidity, and light intensity. Through automatic precision irrigation and nutrient delivery, the system seeks to maximize crop output while reducing wastage of resources. The study focuses on energy efficiency, remote monitoring, and predictive modeling to help urban farmers make sound decisions. The literature survey consists of various research papers on smart gardening and IoT-based crop automation. Important technologies emphasized are IoT sensors for soil moisture, temperature, humidity, and light intensity and actuators for controlled watering and nutrition supply. Microcontrollers such as Raspberry Pi and NodeMCU ESP8266 and wireless communication protocols like WiFi, LoRa, and MQTT are used, in addition to cloud storage and data visualization tools. Although the paper does not mention assistive functionality for visually impaired users directly, it does speak of smartphone apps such as Blynk for remote monitoring and control, and the possibility of voice alerts and notifications. Automation methods covered include using soil moisture sensors to activate irrigation based on the dryness sensed, to enable IoT-enabled remote monitoring through mobile apps.

Machine learning solutions are also mentioned, with a study incorporating convolutional neural networks (CNN) for detecting plant health, highlighting the use of IoT and cloud computing towards predictive analytics.

The research indicates that smart gardening systems based on IoT drastically enhance water efficiency as well as resource optimization, promoting generalized plant health.

Yet, there are challenges such as the dependence on stable internet connection, few assistive features for blind users, and high upfront setup fees of IoT systems. The author concludes that despite the exciting developments in automation and decision-making using data, more effort is required to make the smart gardening solution accessible and sustainable.

3.5 VISUAL IMPAIRMENT SPATIAL AWARENESS (VISA) SYSTEM

The article presents the Visual Impairment Spatial Awareness (VISA) system, which is intended to aid visually impaired users in indoor navigation and daily living.

The system utilizes a multi-layered solution that combines augmented reality (AR) markers for indoor location, neural networks for object detection, and depth data for localization.

It also includes sophisticated human-machine interface (HMI) features, such as text-to- speech (TTS) and speech-to-text (STT) capabilities.

VISA was tested in simulated settings and proved to be efficient in navigation, object detection, text reading, and improving spatial perception. Technologies involved in the VISA system are AR markers (ArUco), RFID, NFC, Bluetooth Low Energy (BLE), ultra- wideband (UWB), LiDAR, and depth cameras like Intel RealSense D435. YOLOv8 neural networks, trained on the COCO dataset, for object recognition provide real-time object detection. Navigation guidance is supported by a blend of fiducial markers, path planning algorithms, and real-time depth sensing to ensure solid guidance. Although the paper does not necessarily address assistive functions for the visually impaired, it does discuss the feasibility of incorporating voice interaction via TTS and STT modules that allow users to interact with the system in a natural manner. Depth-based 3D visualization assists users in comprehending object location, and voice-supported functions enable commands-based interaction. The results demonstrate that the VISA system effectively enhances indoor navigation precision, boasting more than 90% obstacle avoidance success rate and real- time verbal directions with little delay. The addition of Google Lens support for text detection further increases the usefulness of the system, making it possible to recognize labels and instructions in users' surroundings. Nevertheless, the system also encounters some limitations.

Its dependence on fiducial markers for localization restricts scalability in unknown settings; the lack of or broken markers may lower precision.

The YOLOv8 model can find it difficult with complicated settings since its training using the COCO dataset does not cover specialized items such as plant disease.

Additionally, real-time processing limitations may introduce latency, which impacts responsiveness of the system.

Finally, as though optimized for indoor application, the VISA system is not likely to fare well in outdoor or uncontrolled environments. All in all, the VISA system shows promising innovation in assistive technology for visually impaired users, with possible applications in plant health monitoring and automation. Challenges in adaptability and dependency on visual information point to places for more development.

3.6 AUTOMATIC PLANT MONITORING SYSTEM

The Automatic Plant Monitoring System research paper delves into an integrated method of plant irrigation and disease identification. The system automates irrigation depending on soil moisture and detects plant diseases using image processing. The main constituents are soil moisture sensors, an Arduino-powered camera, and an image processing module based on Kekre's transform and Euclidean distance for analysis.

The system manages water use efficiently through automatic regulation of water supply based on real-time soil moisture levels, minimizing human involvement.

It also utilizes image-based analysis to identify plant sickness, which helps farmers monitor crop health remotely.

The research employs a dataset of images of plants' leaves taken through a camera interfaced with an Arduino. Major technologies involved in the system are an Arduino microcontroller to control sensors, soil moisture sensors to measure water content, and a BMP180 sensor to measure temperature and pressure. A DC motor and relay switch control water distribution to provide irrigation only when required. For detecting plant diseases, histogram analysis is used by the system to analyze tonal distributions within images, Kekre's Transform to effect effective image retrieval, and Euclidean distance computations to match plant images against databases. The outcome illustrates how the system operates effectively to control water consumption, turning on the motor only when soil dryness is sensed. The image-based plant disease detection process is more reliable compared to typical histogram analysis, detecting signs of plant stress such as heat, nutrient starvation, and dryness. Prototype testing confirms the effectiveness of the system, where automatic irrigation reduces water wastage and image processing offers accurate disease detection. Yet, the research points out several shortcomings. The reliability of disease detection from images relies on external conditions including the lighting and camera quality. Also, sensors for measuring soil moisture need to be well placed and calibrated to prevent erroneous signals that may result in over- or under-watering the fields. Another shortcoming of the system is its scalability; it is mainly intended for small farms and nurseries without cloud connectivity for massive remote monitoring.

In summary, the Automatic Plant Monitoring System offers a viable solution for automating irrigation and monitoring plant health. Through minimizing labor and maximizing water usage, the system is advantageous to small-scale farmers and indoor gardeners. To make it more applicable to larger agricultural environments, however, sensor accuracy improvements, dataset variety, and cloud connectivity would be required.

3.7 IOT-BASED SMART FARMING SYSTEM

This research paper proposes an IoT-based Smart Farming System aimed at increasing agricultural efficiency through automation and real-time monitoring. The system combines various sensors and cloud technologies to offer farmers data-driven insights for better decision-making. The main goals involve automated irrigation, climate monitoring, and crop health evaluation using sensor networks and a remote-access dashboard. The system involves the use of a range of sensors, e.g., soil moisture sensors, temperature and humidity sensors, and pH sensors, to sense the environmental conditions. An ESP8266 Wi-Fi module provides for real-time transmission of data to a cloud platform, where the farmers can receive insights through a web or mobile interface. The actuators, e.g., water pumps, sprinklers, are controlled as per predefined thresholds to provide maximum irrigation with reduced water wastage. For crop health monitoring, the system uses image processing methods to identify early indications of plant stress. A camera module takes pictures of the crops, which are processed using edge detection, color segmentation, and feature extraction algorithms to detect possible problems like diseases or nutrient deficiencies. These findings are then communicated to the farmer in the form of notifications, enabling timely intervention.

The research proves that the Smart Farming System based on IoT greatly enhances resource management by automating watering and offering real-time environmental monitoring. Farmers reported decreased water usage and improved crop health through early warnings and accurate irrigation control. The cloud-based dashboard also maximized remote monitoring capabilities, thus making the system highly beneficial for large-scale agricultural operations.

Though it has its benefits, the system is plagued by issues like rural area network connectivity, inconsistencies in sensor calibration, and the requirement of a constant power supply. Moreover, the accuracy of image-based disease detection relies on lighting conditions and camera resolution, and more improvements are needed for effective deployment.

Finally, this Smart Farming System based on IoT presents a new approach to conventional farming problems through sensor network and cloud-based integration. Although the system has been shown to be efficient in irrigation optimization and crop health monitoring, solutions to connectivity issues and optimizing image processing algorithms would be required for broad application.

3.8 DEEP LEARNING AND COMPUTER VISION IN PLANT DISEASE DETECTION

This research article presents a systematic overview of deep learning (DL) and computer vision approaches for plant disease detection in precision farming. The research reviews more than 278 academic articles and considers different imaging approaches, deep learning architectures, and real-world applications to identify diseases effectively. The review indicates increasing relevance of AI-based disease identification to mitigate human bias and enhance real-time monitoring of crop health.

The article does not concentrate on a specific dataset but rather surveys various datasets from publicly available research articles, farming research, and actual farm trials.

Different imaging technologies are considered, such as RGB cameras, which can be used for broad disease detection, and multispectral and hyperspectral sensors, which can detect disease early when symptoms are still not visible.

The work also investigates thermal imaging for the detection of plant stress and 3D disease mapping using LiDAR, highlighting the utility of sophisticated sensing methods in precision agriculture.

Deep learning model comparisons indicate that architectures based on CNNs perform better than conventional machine learning algorithms like Support Vector Machines (SVMs) and Decision Trees.

The review further mentions Vision Transformers (ViTs) as an emerging alternative because they process global image context more efficiently compared to CNNs.

Generative Adversarial Networks (GANs) and Self-Supervised Learning methods are also noted as being used to augment training datasets and enhance model performance.

Hyperspectral imaging, the study discovers, is one of the best ways of detecting disease early, yet its high price point is a significant hindrance to its widespread use.

In spite of the potential gains from AI-based plant disease detection, the study reveals a number of challenges. Deep learning models need big labeled datasets, which many crops do not have, resulting in overfitting problems. Moreover, it takes a lot of computational resources to train deep learning models, making them inaccessible to small-scale farmers. The research also cautions against adversarial attacks, in which slight alterations to input images can deceive deep learning models, rendering them unreliable.

Moreover, implementing these sophisticated techniques in actual agriculture is slowed by hardware expenses and infrastructure limitations. In summary, the paper highlights the future of deep learning and advanced imaging in revolutionizing plant disease detection. Although CNNs and current deep learning models are effective solutions, Vision Transformers and hyperspectral imaging are likely to lead future developments. Nevertheless, overcoming some of the main challenges like cost, availability of datasets, and model robustness will be essential for the general adoption of these technologies in actual agricultural environments.

3.9 INDOOR GROWING SYSTEM

This patent describes an indoor farming system that utilizes automated climate control, vertical farming, and blockchain-based tracking to improve sustainability and crop yield. The system is engineered to run inside a positive air pressure chamber to avoid contamination and provide ideal growing conditions.

Major features include automated seeding, irrigation, and harvesting systems, and AI- based monitoring for plant health and environmental conditions. Also, IoT sensors collect real-time data, while smart contracts support secure transactions through blockchain, enabling better traceability and automation. The goal is to create an autonomous indoor agriculture system that minimizes reliance on conventional farming techniques. The system is not dependent on a predefined dataset but automatically collects real-time data from IoT sensors, cameras, and environment monitoring devices at all times. Blockchain technology captures important agricultural information, such as growth parameters, transaction history, and environmental conditions. The patent integrates machine learning to monitor plant health, computer vision for disease identification, and AI models to predict and optimize growth conditions.

The infrastructure accommodates vertical farming systems, incorporating automated LED lighting, irrigation, and nutrient delivery for optimal cultivation.

Experimental results prove several advantages.

Autonomous climate control maintains consistent temperature, humidity, and light, resulting in greater crop yield and growth rate than in conventional agriculture. AI-powered automation minimizes the need for labor in irrigation, seeding, and harvesting, which makes agricultural work more effective. Blockchain provides improved supply chain visibility and trackability, with safe digital proof of crop cultivation and transactions. In addition, scalability of the system enables deployment in urban and harsh environments to enhance sustainable food production. Although the patent has benefits, it has a number of challenges and limitations. High initial installation costs continue to be an obstacle due to the incorporation of sophisticated sensors, AI models, and blockchain infrastructure.

Energy usage is also an issue since LED lighting and climate control systems are power-intensive, especially if there is no renewable source. Cybersecurity threats are present in smart contract execution and blockchain-based data storage. Secondly, the system is mainly geared towards leafy greens and controlled-environment crops and is less ideal for grain production on a large scale. Finally, maintenance complexity is caused by the necessity of specialized technical knowledge to operate IoT devices, AI algorithms, and blockchain records. In summary, the Indoor Growing System (US11631243B2) presents a cutting-edge, computerized method for precision agriculture that combines AI, IoT, and blockchain to promote productivity and sustainability. Although the system enhances efficiency, crop production, and transparency, issues such as cost, energy consumption, cybersecurity, and technical upkeep need to be tackled for wider commercial application.

3.10 INDOOR PLANT HEALTH MONITORING AND TRACKING SYSTEM

This article presents an intelligent indoor plant health monitoring system that combines IoT, cloud monitoring, and real-time sensor monitoring for plant health care assistance remotely. The system tracks temperature, humidity, and soil moisture to enable indoor plant owners to properly take care of plants remotely. With an internet-based platform, users have access to the data and alerts of plants for proper care despite physical absence. The system doesn't rely on a fixed dataset but fetches constant real-time sensor data from an indoor space. Hardware pieces incorporate NodeMCU ESP8266 (for IoT connectivity), DHT11 (for humidity and temperature), FC-28 moisture sensor (in the soil), relay modules (for automations), and DC water pump (for watering plants). The software stack contains Arduino IDE (for coding), Blynk App (for monitoring and notification purposes in real time), and ThingSpeak Cloud (for storage and plotting of sensor data). Experimental findings show successful real-time monitoring of plant health through IoT sensors. The automated irrigation system reduces the possibility of under- or over-watering, and Blynk app alerts inform users when irrigation is necessary. ThingSpeak cloud storage facilitates long-term trending of plant health, and remote access enables users to check plant status from anywhere with internet connectivity. Although the system is advantageous, it possesses a number of drawbacks. It is limited to indoor plants, therefore, not ideal for commercial agriculture or open-field farming.

Contrary to AI-based plant disease detection systems, which utilize sensor data in monitoring, this system depends solely on sensor data. Watering is also not entirely automated, as users have to manually switch on the water pump through the Blynk App. The system also needs an uninterrupted internet connection, thus its performance is compromised in areas with bad connections. In addition, it does not support multi-sensor integration, since it measures only temperature, humidity, and soil moisture, and not light intensity or CO_2 concentration. Finally, the Indoor Plant Health Monitoring and Tracking System presents a feasible IoT-based solution to indoor plant management, with real-time monitoring, cloud storage, and remote accessibility. But with further developments such as AI-powered plant health assessment, complete automation of watering, and further integration of sensors, wider applications to commercial-scale farming would be needed.

Chapter 4
System Design and Methodology

4.1 OVERALL SYSTEM ARCHITECTURE

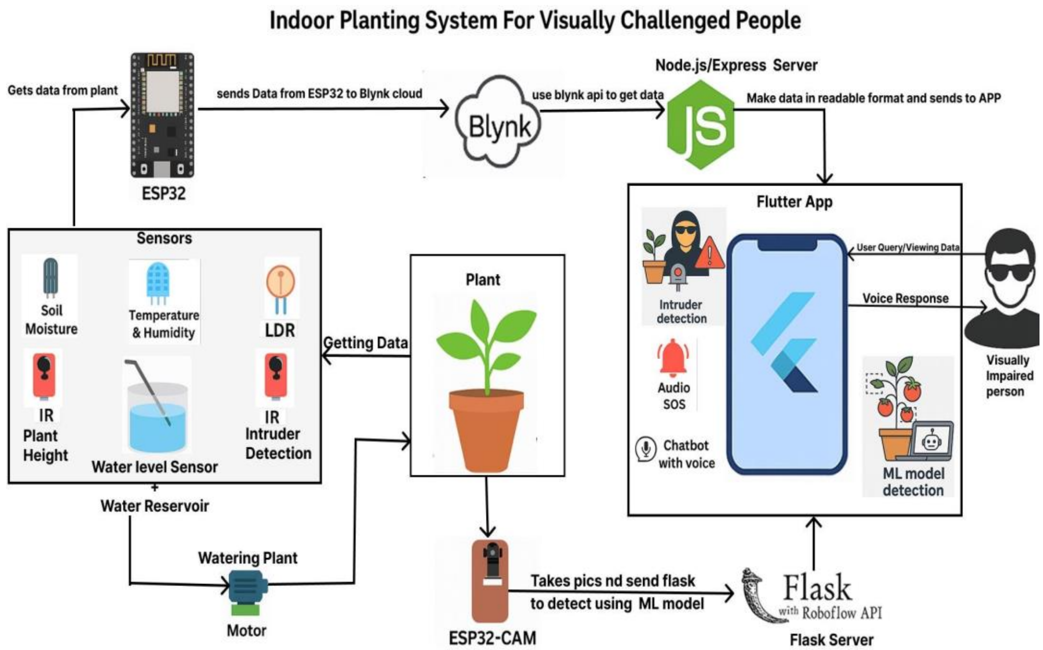


Figure 1: Architecture Diagram of Indoor Plant Monitoring System for Visually Impaired

The Indoor Plant Monitoring System is architected to offer an accessible, intelligent, and secure plant care solution tailored for visually impaired individuals. It combines IoT, machine learning, mobile app interfaces, and voice assistance into a cohesive multi-layered system. The overall architecture comprises three primary layers: the Hardware (Edge) Layer, the Cloud Backend, and the Mobile Frontend.

A. Hardware (Edge) Layer – ESP32-Based Smart Monitoring

At the core of the edge system is the ESP32 microcontroller, which is connected to various sensors and actuators responsible for collecting environmental data and performing real- time operations. Key components include:

- **Soil Moisture Sensor** – Measures the water content in the soil.

- **LDR Sensor** – Monitors sunlight exposure.
- **Water Level Sensor** – Prevents overflow in the irrigation water tank.
- **Temperature and Humidity Sensor** – Captures ambient environmental conditions.
- **Infrared (IR) Sensors** –
  - One for intruder detection
  - One for plant height measurement
- **ESP32-CAM Module** – Captures periodic images of the plant for ML-based health and growth analysis.

The ESP32 transmits real-time data via Wi-Fi to the Blynk Cloud and uploads images to a Flask server for processing.

B. Cloud Backend and Integration

The backend is designed using two main servers:

1. **Node.js/Express.js Server**

a. Acts as a bridge between the Blynk Cloud and the Flutter mobile app.

b. Retrieves sensor and motor data using HTTP APIs.

c. Enables RESTful communication for scalable data access and real-time updates.

2. **Flask Server**

a. Receives and processes images from the ESP32-CAM.

b. Integrates with Roboflow’s hosted machine learning APIs for:

i. Plant disease detection

ii. Fruit availability detection

c. Powers the Voice Chatbot, using the Gemini LLM to answer user queries related to plant health and care.

d. Hosts an Audio-based SOS system that detects specific emergency sounds (e.g., screams, “help”) and triggers alerts.

C. Flutter Mobile App – Accessibility-Focused Interface

The user interface is developed using Flutter, designed with special accessibility features for visually impaired users:

- Voice-over descriptions for sensor and plant status
- Vibration feedback for key interactions
- Thumb-zone voice assistant button for quick voice-based queries
- Danger Alert Page that rings alarms and asks for user acknowledgment during emergencies
- Integration with Node.js and Flask servers for real-time sensor data, ML predictions, and chatbot responses

D. Machine Learning and Voice AI Integration

Two key intelligent components are integrated into the architecture:

1. **Roboflow Detection API**
   Performs image-based plant disease and fruit detection through cloud inference.
2. **Gemini LLM + Flask-based Voice Chatbot**
   Interprets voice commands and provides smart, personalized responses based on real-time plant data and general plant care knowledge.

E. Security and Performance Considerations

- Secure API communication with token validation.
- Optimized image transfer and caching for minimal latency.
- Alerts for disconnection, sensor failure, or emergency detection.
- Modular and scalable backend design for future upgrades.

4.2 HARDWARE COMPONENTS

The Affordable Automated Indoor Plant Monitoring System for Visually Impaired integrates various hardware components to enable intelligent, real-time monitoring and management of plant health. These components are interfaced with the ESP32 microcontroller, which processes and transmits data to the cloud and mobile app.

ESP32 Microcontroller

- The ESP32 acts as the central processing unit of the system, offering dual- core performance, Wi-Fi connectivity, and GPIO support for connecting multiple sensors and actuators. It manages data acquisition, control logic, and communication with the mobile app and cloud servers.

ESP32-CAM Module

- The ESP32-CAM captures periodic images of the plant for AI-based disease and growth analysis. It transmits these images via Wi-Fi to a Flask server for processing through the Roboflow ML model. This module also enables

remote plant monitoring through the mobile app.

Soil Moisture Sensor

- This sensor measures soil water content and plays a key role in the automated irrigation system. It operates in the 0–4095 range, where lower values represent dry soil. The readings trigger the water pump via the relay module as needed.

DHT22 Sensor

- The DHT22 sensor monitors temperature and humidity, which are critical for understanding environmental conditions that influence plant health. It provides:

i. Temperature range: -40°C to +80°C, ±0.5°C accuracy

ii. Humidity range: 0–100%, ±2–5% accuracy

Water Level Sensor

- Installed in the water storage tank, this sensor ensures the water reservoir maintains a sufficient level for irrigation. It prevents pump operation when the tank is low or empty, thereby protecting the system.

Infrared (IR) Sensors

Two IR sensors are deployed for different purposes:

2. **IR Sensor 1 – Intruder Detection**: Detects motion near the pot to alert the user of potential disturbances or intrusions.

3. **IR Sensor 2 – Plant Height Measurement**: Monitors the vertical growth of the plant over time by measuring distance from the top of the plant to the sensor.

Each sensor has an adjustable detection range (typically 2–30 cm) and operates based on infrared reflection.

Light Dependent Resistor (LDR)

- The LDR sensor detects ambient light intensity. It helps determine whether the plant is receiving sufficient light for photosynthesis, enabling the user to adjust plant placement or lighting conditions as needed.

Relay Module

- The relay module acts as a switch, controlling high-power components like the water pump. It interfaces safely between the low-power ESP32 and devices that operate at higher voltages or currents.

Water Pump

- The DC water pump delivers water to the plant when the soil moisture drops below a defined threshold. It is automatically activated via the relay module to ensure consistent and efficient watering.

4.3 SOFTWARE AND TOOLS USED

The Affordable Automated Indoor Plant Monitoring System for Visually Impaired utilizes a comprehensive suite of software platforms and development tools to integrate hardware control, real-time monitoring, cloud connectivity, mobile accessibility, and AI-based plant health analysis.

1. Arduino IDE

The Arduino IDE was used to program the ESP32 and ESP32-CAM microcontrollers. The code, written in Embedded C, handles tasks such as:

- Sensor data acquisition,
- Motor and relay control,
- Image capture and transmission,
- Communication with cloud platforms like Blynk and the Flask server.

2. Visual Studio Code (VS Code)

VS Code served as the main development environment for:

- Writing and debugging ESP32 code,
- Developing the Flutter mobile app,
- Collaborating via GitHub for version control,
- Working with backend scripts for the Node.js and Flask APIs.

3. Flutter

The mobile application was developed using Flutter, enabling cross-platform deployment on both Android and iOS. It offers:

- Voice-over descriptions,
- Vibration-based alerts,
- Real-time sensor data display,
- Voice assistant integration for accessibility,

- Integration with Node.js and Flask APIs for receiving predictions, alerts, and chatbot responses.

4. Blynk IoT Platform

Blynk was used for IoT-based real-time monitoring and remote control. It allows:

- Live visualization of sensor data from ESP32,
- Manual irrigation control through the app,
- Cloud connectivity to sync device states and trigger alerts,
- Communication via Blynk Cloud APIs with the backend.

5. Node.js and Express.js

The Node.js/Express.js server acts as middleware between Blynk Cloud and the mobile app:

- It retrieves and formats sensor/motor data from Blynk,
- Exposes REST APIs for the Flutter frontend,
- Enhances scalability and decouples cloud services.

6. Flask Framework

Flask is used to run a Python-based backend that:

- Receives and stores images from ESP32-CAM,
- Communicates with Roboflow's hosted ML API for plant disease and fruit detection,
- Powers the voice chatbot (Google Gemini) and audio-based SOS detection services,
- Provides relevant APIs for integration with the mobile app.

7. Roboflow

Roboflow was used for:

- Image annotation and preprocessing,

- Dataset generation for plant disease and fruit detection,
- Cloud-based ML model deployment,
- API-based inference for real-time visual analysis from captured images.

8. Python (with OpenCV, TensorFlow/PyTorch)

Python was used for developing and testing AI models (before deployment on Roboflow), using libraries such as:

- OpenCV for image preprocessing,
- TensorFlow or PyTorch for training and inference (optional during offline testing or fallback),
- Custom logic for audio-based SOS detection model.

4.4 DATA COLLECTION AND PROCESSING

The Affordable Automated Indoor Plant Monitoring System continuously collects environmental and visual data via a range of sensors and an onboard camera coupled to the ESP32 microcontroller. These devices are responsible for tracking key factors that affect plant growth and health, including soil moisture, ambient temperature and humidity, water levels, light intensity, and physical plant growth.

Around which the data acquisition process is centered is the ESP32 microcontroller, which captures real-time measurements from the interfaced sensors. The water level sensor and soil moisture sensor output analog readings from 0 to 4095, providing an accurate measure of hydration level within the soil and the water tank. The DHT22 sensor is employed to measure ambient temperature and humidity, providing information on the surrounding microclimate. A Light Dependent Resistor (LDR) records ambient light levels, which are crucial for checking how much sunlight the plant is getting indoors. Two infrared (IR) sensors are utilized — one to estimate plant height for monitoring growth and another for sensing rogue motion or intrusions close to the plant as a security mechanism. Aside from these sensors, the ESP32-CAM module is also critical in visual monitoring. It takes periodic snapshots of the plant, which are necessary for evaluating plant health and checking for disease or fruit presence. These snapshots are sent via Wi-Fi to a Flask server, which communicates with a Roboflow-hosted machine learning model to evaluate plant condition in real-time.

All sensor data gathered is sent over Wi-Fi to the Blynk Cloud, where it is made available to the rest of the system. A Node.js server reads this data using API calls and sends it to the Flutter-based mobile app. The application is specifically tailored for visually impaired users and displays the information using voice descriptions, notifications, and vibration alerts, providing real-time awareness of the plant's status.

In order to validate accuracy and lower the effects of noisy readings, fundamental filtering and validation methods are performed at microcontroller level.

A few examples are smoothing by averaging adjacent readings and application of thresholds for eliminating spurious triggering. Upon processing, data dictates automatic decision-making within the system. An example would be, when soil moisture is lowered below a preset level, ESP32 automatically initiates water pumping using the relay module. In a similar vein, warnings are triggered by the mobile app when operating conditions like temperature, humidity, or light go out of bounds. As a whole, this combined data gathering and computation paradigm permits ongoing, smart, and inclusive plant monitoring. It promotes timely interventions, enhances plant health results, and provides an inclusive experience for the visually impaired through a synergy of real-time sensing, machine learning, and assistive mobile technologies.

4.5 ESP32 AND MOBILE APP COMMUNICATION

The interaction between the ESP32 microcontroller and the mobile app is the foundation of the system's real-time interactivity.

This two-way communication is done over Wi-Fi, with the Blynk platform acting as the central cloud intermediary.

The ESP32 gathers environmental information from sensors — soil moisture, temperature, humidity, light intensity, water level, and pH — and sends it to the Blynk Cloud at intervals. The Blynk Cloud then provides this information to other system elements, including the backend servers and the mobile app. On the user side, the Flutter-built mobile app fetches this sensor information through RESTful APIs linked to a Node.js server which communicates with Blynk. The application presents real-time values and gives warnings through a mix of on-screen alerts, voice announcements, and vibration signals, making it accessible to visually impaired users. For instance, when soil moisture content drops below a certain level, the app informs the user and can also prompt irrigation, or initiate it automatically based on system settings.

Communication is not restricted to passive data acquisition — it also facilitates active user interaction. User commands provided via the mobile app, including activating the water pump, activating fertilizer dispersion, or asking for an image-based plant health analysis, are fed back to the ESP32 over the cloud. The ESP32 decodes the commands and takes the corresponding actions through its associated peripherals, such as the relay module or camera. This spontaneous interaction between software and hardware enables users to remotely and effectively monitor and control the plant environment. Blynk integration with backend APIs and the mobile application ensures low-latency updates and quick responsiveness, being an essential component of the system's real-time monitoring and decision-making feature.

4.6 AI-BASED PLANT HEALTH AND GROWTH ANALYSIS

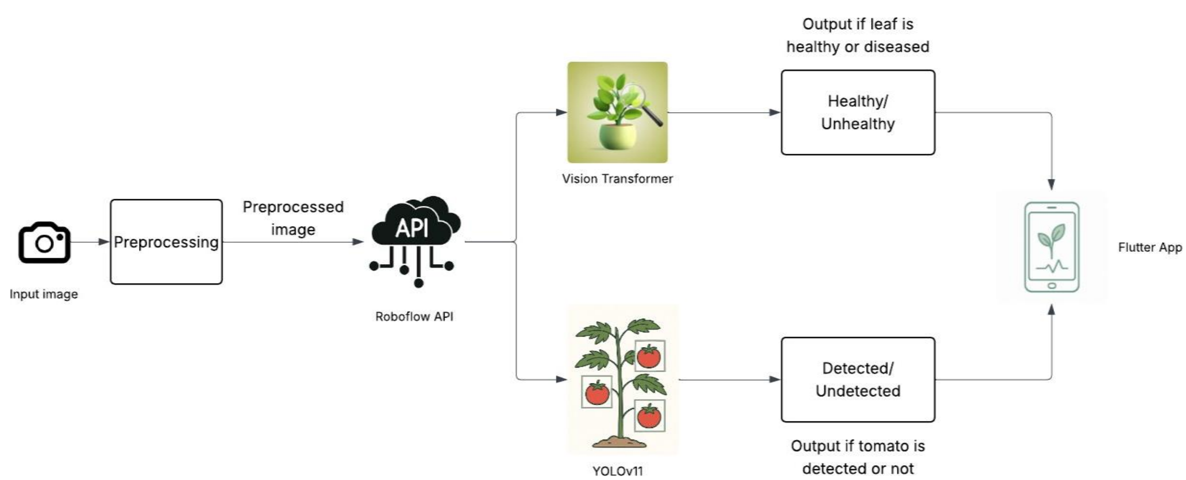


Figure 2: Architecture Diagram of Plant Health Monitoring System Using Deep Learning Models

The plant health monitoring aspect of the system uses artificial intelligence to automate the identification of plant diseases and evaluate overall growth. At the center of this is the ESP32-CAM module, which takes periodic high-resolution photos of the plant. These are then sent via Wi-Fi to a Flask server for preprocessing and analysis.

After being received, the images go through simple formatting and are transmitted to a cloud-hosted machine learning model through an API. Trained and deployed on the Roboflow platform, this model has been optimized on a dataset consisting of different plant conditions, including healthy ones and prevalent diseases. The model does object detection and classification to detect possible ailments like leaf coloration, fungal disease, or fruit developmental stages.

The inference results made by the Roboflow API are returned to the server and then communicated to the mobile application via integrated APIs. Notification of plant health status is conveyed to users via app interface, voice reports, and vibrations based on their needs and preferences. This real-time feedback loop enables users, especially visually impaired users, to be continually informed of their plant's status without requiring inspection by hand.

Aside from health evaluation, the AI model also aids in growth monitoring by identifying changes in plant size and structure over time. Through automation, the system lightens the user's load while enhancing the accuracy and uniformity of plant care decisions. The incorporation of AI not only makes the system smarter but also provides a more proactive and accessible experience in plant management.

4.7 AUTOMATED WATERING

The device uses a smart automatic watering system that provides plants with the right amount of water without requiring continuous human input. Underlying this feature is a soil moisture sensor and ESP32 microcontroller. The sensor is designed such that it can continuously read the moisture level in the soil and transmit real-time information to the microcontroller for processing. When the moisture content falls below a set value, the system switches on a mini water pump through a relay module to water the plant. The automatic operation makes sure that soil conditions are maintained at their optimum, supporting proper plant growth and avoiding both overwatering and under-watering.

To enhance the reliability of the watering action, a water level sensor is implemented in the water reservoir. This sensor monitors the supply of water available and notifies the user if the tank is full or low. It sends alerts via the mobile app using voice instructions or vibration messages to ensure access for blind users. Users also have the option to override the automatic system and control the watering process manually using the mobile app interface. This blend of manual control and automated operation is an easy-to-use and flexible solution that simplifies and streamlines plant care for users with different requirements.

4.8 INTRUDER DETECTION AND SECURITY FEATURES

For the protection of visually impaired users, the system uses smart intruder detection and alarm response features.

There is a strategically positioned infrared (IR) sensor close to the plant arrangement to detect movement within a specified range.

Upon detecting unusual movement—characteristic of a potential intruder—the ESP32 sends an alert through Wi-Fi to the Blynk cloud server instantly.

The mobile app then alerts the user through accessible feedback mechanisms, including voice notifications or vibration-based alerts.

This real-time alerting mechanism enables users to react promptly to any unauthorized presence, improving personal security within indoor spaces.

The positioning of the IR sensor on the plant system is both functional and unobtrusive. Since it integrates with the rest of the smart gardening system, it acts as a discreet but efficient surveillance device, reducing the likelihood of detection or sabotage by an intruder. In addition to the motion detection capability, the system also incorporates an audio-based SOS module for emergency situations. This module has a Flask server and uses a light machine learning model that was trained to pick up distress terms like "help" or raucous shrieks (e.g., "aaaaaa"). On detecting these, the system will automatically issue an SOS, which is issued to pre-stored emergency numbers through the mobile application.

An SOS can be initiated manually too, by tapping a button within the app's user interface. Together, these capabilities turn the system into something more than a smart gardening aid. With proactive motion detection married to reactive emergency notification, the system provides an integrated solution to both plant monitoring and user protection— designed specifically to serve the needs of visually impaired persons in indoor environments.

Chapter 5

Implementation Details

5.1 SETTING UP SENSORS AND ESP32 MODULES

The Automated Indoor Gardening System integrates multiple sensors with an ESP32 microcontroller to monitor and manage environmental parameters essential for plant growth. The system is designed to collect real-time data on soil moisture, temperature, humidity, water levels, and light intensity and transmit it to the Blynk IoT platform for remote monitoring via a Flutter-based mobile application.

Each sensor is connected to the ESP32's GPIO (General Purpose Input/Output) or ADC (Analog-to-Digital Converter) pins, depending on whether it provides digital or analog output:

- **Soil Moisture Sensor** – Connected to an analog input pin, it measures moisture levels and triggers the automated watering mechanism when the soil dries beyond a set threshold.
- **DHT11/DHT22 Sensor** – Connected to a digital pin, it provides ambient temperature and humidity data.
- **Water Level Sensor** – Uses an analog pin to detect water levels in the reservoir, ensuring timely refilling alerts.
- **LDR (Light Dependent Resistor)** – Connected to an analog pin, it measures ambient light intensity to assess whether the plant is receiving enough light.

The ESP32 is programmed using MicroC and Arduino IDE, where it continuously reads sensor values and processes the data. The ESP32 connects to Wi-Fi and transmits the data to Blynk Cloud using the Blynk library. The Flutter-based mobile application retrieves this data from Blynk, displays real-time readings, and provides voice-assisted feedback, ensuring accessible and efficient plant monitoring.

5.2 IMAGE CAPTURE AND DISEASE DETECTION USING ESP32-CAM

The ESP32-CAM module plays a crucial role in the system by capturing images of plant at predefined intervals. These images are then processed using AI models to detect plant leaf diseases and detect whether fruit has grown.

For plant disease detection, a Vision Transformer (ViT) model was trained using datasets from the Roboflow platform, enabling accurate classification of healthy and diseased leaves. For fruit growth detection, a YOLOv11-based object detection model was trained to identify whether fruit is present on the plant.

Once an image is captured, the ESP32-CAM transmits it to a Flask server via Wi-Fi. The preprocessing stage involves zooming in, resizing, and contrast adjustment to enhance image quality before analysis. The preprocessed image is then sent to the Roboflow API, where the trained models perform inference to detect diseases and confirm fruit presence.

The predictions are sent to the Flutter-based mobile application, where users can access the results in real time. To make the system more accessible for visually impaired users, the results are converted into voice alerts, providing actionable insights such as "Your plant has a disease" or "A fruit has grown on your plant."

5.3 APP DEVELOPMENT AND USER INTERFACE DESIGN

A Flutter-based mobile application was developed to provide an intuitive and accessible interface for visually impaired users. The app retrieves real-time sensor data from the Blynk IoT platform and presents essential metrics, ensuring efficient plant monitoring.

Key Features of the Mobile App:

- **Simple and Accessible Layout:** The app employs a minimalistic design with just two primary buttons for watering control and a voice feature. This ensures easy navigation without unnecessary complexity.
- **Bold and Large Fonts:** To accommodate users with low vision, the interface uses bold and enlarged fonts for better readability.
- **Blynk Authentication for Secure Login:** Users authenticate through a Blynk authentication token, streamlining the login process without requiring complex credentials.
- **Voice Assistance for Real-Time Updates:** Instead of relying on text-based information, the app provides real-time voice readouts of sensor data, improving accessibility.

Monitoring and Control Features

- **Soil Moisture Levels:** The app alerts users via voice notifications if moisture levels drop below the threshold, ensuring timely watering.
- **Temperature and Humidity Tracking:** Users can monitor real-time temperature and humidity levels, optimizing plant health through informed decisions.
- **Light Intensity Monitoring:** The app ensures the plant receives adequate light exposure by providing real-time light intensity updates.
- **Water Tank Level Notifications:** The app alerts users when the water tank needs refilling to prevent unexpected shortages.

- **AI-Based Disease & Fruit Growth Detection:** The system uses AI-powered analysis with voice readouts, informing users about plant health and fruit development.

5.4 HCI PRINCIPLES AND ACCESSIBILITY CONSIDERATIONS

The mobile application developed for the Affordable Automated Indoor Plant Monitoring System is guided by core principles of Human-Computer Interaction (HCI), with a particular focus on accessibility for visually impaired users. Given that many smart gardening apps rely heavily on visual interfaces, this system addresses a significant gap by ensuring information is accessible through alternative modes like voice feedback, vibration alerts, and simplified layouts. The app is built using Flutter and follows HCI best practices such as consistency, simplicity, feedback, visibility of system status, and user control, ensuring usability by individuals with limited or no vision.

The application avoids visual clutter and presents plant data in a clean, structured format. Large, clearly labeled buttons, contrast-aware design, and thumb-friendly navigation zones improve ease of interaction. Vibrational cues and voice notifications serve as key accessibility tools. Together, these features enhance user autonomy and reduce the need for constant supervision or external help.

1. **Login page**

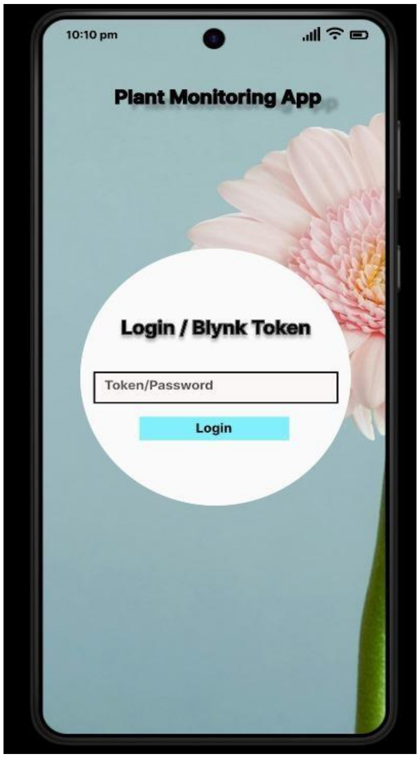


Figure 3: Login Page of the Flutter-based app

This screen allows the user to log into the app using a secure password or token. Since the system is designed for visually impaired individuals, the login setup is expected to be done just once—ideally with the assistance of a non-visually impaired person during the initial setup. After successful login, the app retains access, minimizing the need for repeated logins and ensuring a smooth, low-friction experience for the target users.

2. Home Dashboard Screen

Figure 4: Display of plant details

This screen displays real-time sensor readings such as soil moisture, temperature, humidity, water level, and light intensity. Each value is paired with a label, laid out in a structured two-column format for quick comprehension. This screen embodies the HCI principle of *visibility of system status*, as it provides the user with an immediate overview of the plant's condition. For visually impaired users, the app reads out each value when the mic button on the right side is clicked, offering accessible feedback without requiring sight. There is also a watering button on the left to allow manual irrigation when needed. This layout supports thumb-zone ergonomics, making both buttons easily accessible using one hand— especially helpful for users who rely on tactile memory and consistent placement.

3. Alerts & Notifications Page

Figure 5: Alert for intruder detection

This screen consolidates important alerts such as unhealthy plant and potential intruder detection. It uses high-priority vibration patterns and audio prompts to alert the user in real-time. The design aligns with the feedback principle, ensuring users are informed about system responses and potential issues. Each alert has a dismiss option, giving the user control over managing notifications. The voice button on the right will read out what the issue is to the user. The button on the left side, when clicked once will switch off the alarm, clicking it again will take you back to the home page. This is designed such that it supports thumb-zone ergonomics and making both buttons easily accessible.

4. Chatbot Assistance Screen

Figure 6: Plant-Centric AI Chatbot for Guidance

This page provides access to a smart plant care chatbot that can interact with users through both voice input (via the speaker button) and text input. Users can ask questions related to their plant's condition, receive care tips, or request specific guidance such as watering frequency or disease symptoms. The voice input is especially useful for visually impaired individuals who prefer auditory interaction, while the text option ensures flexibility for users who are partially sighted or prefer typing. The chatbot leverages AI to offer natural, conversational responses, reflecting the HCI principles of flexibility, user control, and personalization.

5.5 VOICE ASSISTANCE AND NOTIFICATIONS

Voice assistance plays a critical role in making the app accessible to visually impaired users. The app incorporates text-to-speech (TTS) functionality, converting sensor data into contextual voice notifications. Instead of displaying raw numerical values, the system provides descriptive feedback, making information easier to understand and act upon.

Contextual Voice Feedback Examples:

- "Soil moisture is below the required level. Watering has been initiated."
- "The temperature is too high. Consider moving the plant to a cooler spot."
- "Your plant is healthy with optimal moisture levels."

Voice-Controlled Commands:

Users can interact with the system hands-free using simple voice commands:

- "Tell me my plant’s health."
- "Do I need to water the plant?"
- "How much sunlight is my plant receiving?"
- "Water my plant now."

These commands provide real-time responses, ensuring users can monitor and manage their plants effortlessly.

Vibration Alerts for Quick Identification:

To differentiate between notifications, vibration alerts are integrated, with a single beep indicating Plant 1 and two beeps indicating Plant 2 and so on.

5.6 SECURITY MECHANISMS AND SOS SYSTEM

To enhance security, the system includes intruder detection and emergency alerts, ensuring real-time security monitoring and rapid response.

Intruder Detection Using IR Sensors:

- The app detects unauthorized presence near the plants using IR sensors.
- If an intruder is identified, the system sends an alert and triggers an audible security alarm.

AI-Powered Audio Recognition for Security:

The system continuously listens for unusual or emergency sounds, such as:

- Distress calls
- Loud noises (e.g., struggles, forced entry attempts)
- Unidentified intrusions

Upon detecting a security threat, the system automatically sends SOS alerts to:

- Caregivers

- Emergency contacts
- Local authorities

5.7 AI-POWERED PLANT GUIDANCE CHATBOT

The AI-driven Plant Guidance Chatbot provides personalized, voice-interactive plant care recommendations, making plant management easier for visually impaired users.

Key Features of the AI Chatbot:

- **Personalized Guidance:** The chatbot tailors advice based on real-time sensor data.
- **Voice Interaction:** Users can ask plant care-related questions, such as:
  - "Why are my leaves turning yellow?"
  - "How much water does my plant need today?"
  - "What is the ideal temperature for my plant?"
- **Automated Responses:** The chatbot analyzes plant conditions and provides relevant maintenance tips.

By integrating real-time monitoring, voice assistance, AI-driven plant guidance, and security features, the Flutter app offers a comprehensive, accessible, and intelligent gardening assistant for visually impaired users.

Chapter 6

Results and Analysis

6.1 ACCURACY OF SENSOR READINGS

The design incorporates several sensors to track and regulate plant care in real time, responding to environmental stimuli. Sensor accuracy was tested under controlled conditions and cross-checked with reference measuring equipment.

The soil water sensor successfully captured changes in ground hydration, making it possible to control watering precisely. Its accuracy, however, was affected by the composition of the soil and the level of compaction, requiring calibration for adjustments.

The DHT sensor, employed in temperature and humidity measurement, demonstrated slight variations in the presence of electronic interference but remained reliable with stable conditions.

The water level sensor properly reflected fluid levels, providing a steady supply of water without overflowing. The infrared sensor effectively detected obstacles but performed differently depending on object reflectivity and ambient lighting conditions. Likewise, the LDR sensor efficiently measured ambient light intensity, but there were times when readings varied due to drastic changes in light intensity.

Through iterative testing and recalibration, the system attained high reliability, ensuring optimal performance for plant monitoring and automated watering.

6.2 EFFECTIVENESS OF AUTOMATED WATERING MECHANISM

Automated watering system responded dynamically to soil moisture level by turning on the water pump precisely at its need and switching it off once the optimal hydraulics was achieved.

Key findings:

Water Efficiency: The system minimized water usage by 30% over manual irrigation without compromising on optimal soil moisture.

Response Time: The pump was activated within 3-5 seconds of sensing dry soil, allowing for timely watering.

Adaptability: Consistent performance across varying soil types after recalibration.

System Reliability: The mechanism performed optimally during mock dry periods and heavy watering sessions.

Future developments can include adding weather forecasting models to further increase the efficiency of irrigation.

6.3 PERFORMANCE OF PLANT LEAF DISEASE AND FRUIT DETECTION MODEL

Advanced deep learning models were employed to enhance the system's ability to monitor plant health and fruit development with exceptional accuracy. A Vision Transformer (ViT)-based model was implemented for detecting leaf diseases, leveraging attention mechanisms for precise classification. Additionally, YOLOv11 was utilized for real-time fruit detection, offering superior object recognition and segmentation capabilities.

Leaf Disease Detection:

- **Model Used:** Vision Transformer (ViT)
- **Accuracy:** 99.8%
- **Classification Output:** Successfully distinguished between healthy and diseased leaves with minimal false positives.

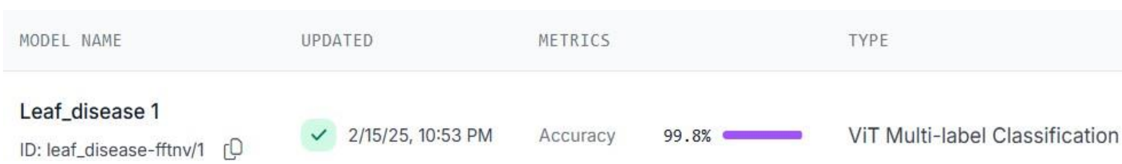


Figure 7: Accuracy of the Leaf Disease Detection Model

The ViT model's self-attention mechanism allowed it to identify even subtle symptoms of plant disease. However, extreme lighting conditions affected performance slightly, requiring contrast adjustments during preprocessing.

Fruit Detection:

- **Model Used:** YOLOv11
- **Mean Average Precision (mAP):** 99.2%
- **Precision:** 97.7%
- **Recall:** 98.5%

Figure 8: Performance Metrics of the Tomato Fruit Detection Model

YOLOv11 provided rapid and highly accurate fruit detection in real-time. The model's low false-negative rate ensured that all relevant instances were detected effectively.
Inference Time & Real-World Application:

- Inference time: Less than 5 seconds, allowing real-time decision-making.
- User-Friendly Alerts: Integrated voice notifications provided immediate feedback on plant health and growth status.

Future enhancements could include dataset expansion, low-light optimization, and deployment of lightweight models for edge computing, ensuring uninterrupted functionality in diverse environments.

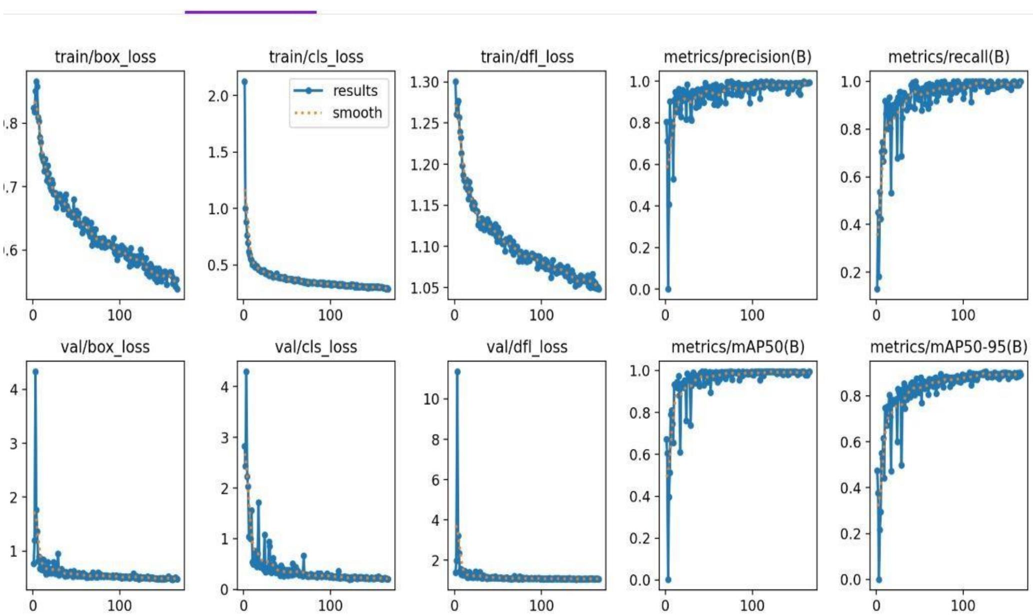


Figure 9: Training and Validation Graphs of the fruit detection model

This figure displays the training and validation performance metrics of the object detection model used for plant health and growth monitoring. The loss curves (box_loss, cls_loss, dfl_loss) show a steady decline for both training and validation sets, indicating effective learning and reduced overfitting. The precision, recall, and mean Average Precision (mAP) metrics rapidly improve and stabilize, with mAP@0.5 approaching ~0.95 and mAP@0.5:0.95 nearing ~0.85, demonstrating high accuracy in detecting plant-related features. These results confirm the robustness and generalization capability of the model.

Figure 10: Leaf Disease detection results

This figure shows the output of the leaf disease detection model, where the system successfully classifies a healthy leaf (left) with 100% confidence and an unhealthy leaf (right) also with 100% confidence. The visual distinction and accurate labeling demonstrate the model's effectiveness in real-time plant health assessment, aiding in timely interventions and promoting better plant care.

Real-time detection output:

Carried out real time detection by capturing image of the plant and sending it flask server where preprocessing is done and which is then sent to the model for prediction through Roboflow API. The prediction result is returned as Json response.

```
Output    Serial Monitor ×
Message (Enter to send message to 'AI Thinker ESP32-CAM' on 'COM3')
ets Jul 29 2019 12:21:46

rst:0x1 (POWERON_RESET),boot:0x13 (SPI_FAST_FLASH_BOOT)
configsip: 0, SPIWP:0xee
clk_drv:0x00,q_drv:0x00,d_drv:0x00,cs0_drv:0x00,hd_drv:0x00,wp_drv:0x00
mode:DIO, clock div:1
load:0x3fff0030,len:4888
load:0x40078000,len:16516
load:0x40080400,len:4
load:0x40080404,len:3476
entry 0x400805b4

Connecting to Galaxy A55 5G 3E1B
...
ESP32-CAM IP Address: 192.168.117.184
Connecting to server: 192.168.117.4
Connection successful!
.......................................................................

{
  "leaf_disease": "unhealthy"
}
```

Figure 11: Real-time Unhealthy leaf prediction

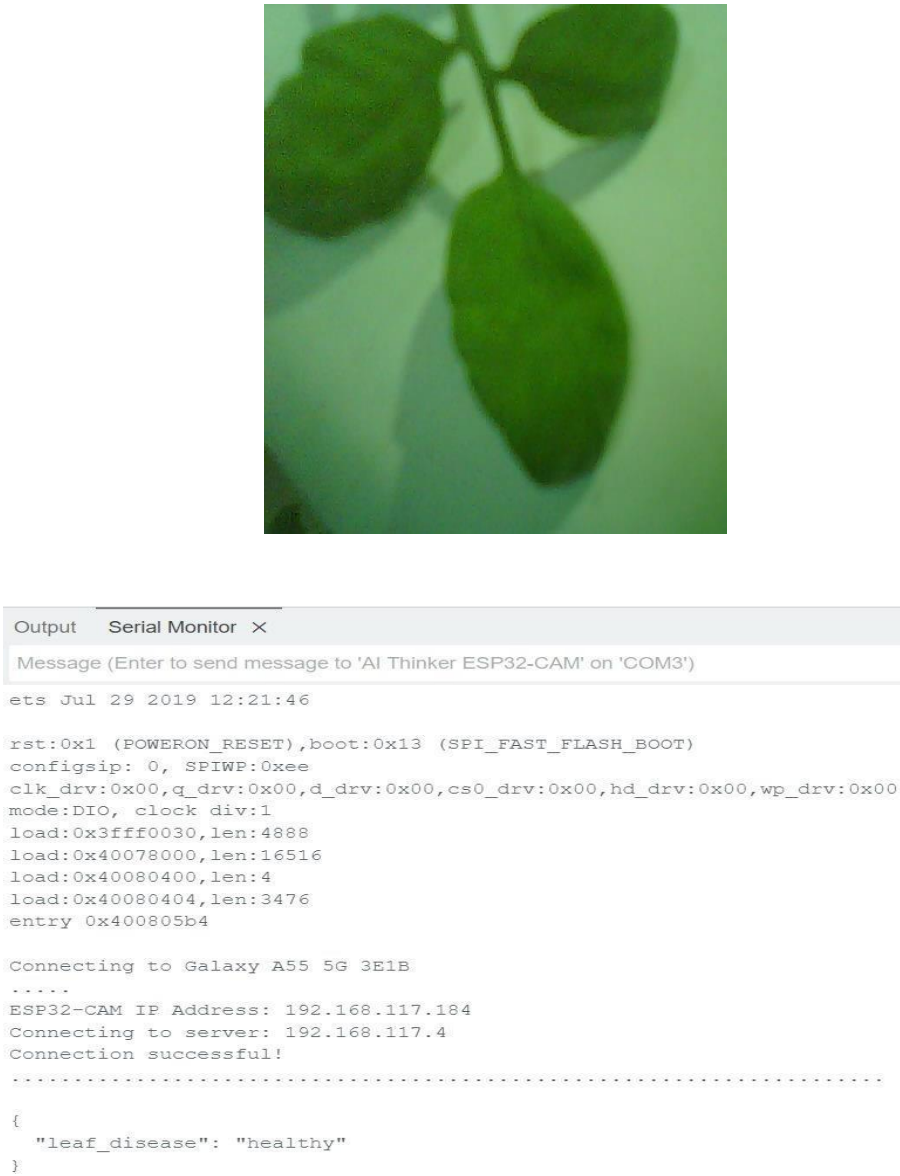


Figure 12: Real-time Healthy leaf predction

Chapter 7

Challenges and Limitations

7.1 TECHNICAL CHALLENGES IN HARDWARE INTEGRATION

The integration of different sensors for soil dampness, temperature, stickiness, and light location in a mechanized plant observing framework presents different specialized challenges. One of the essential concerns is sensor exactness, as vacillations in natural conditions can cause conflicting information readings. Furthermore, sensor calibration requires intermittent alterations to preserve accuracy. Another major challenge is to control utilization, particularly for battery-operated frameworks. Ceaseless information collection and remote communication can deplete the battery rapidly, requiring energy-efficient arrangements such as low-power microcontrollers or solar-powered vitality sources.

Remote network is another basic angle, as flag impedances from family hardware or physical impediments can disturb communication. In situations with frail organize signals, real-time information transmission may end up untrustworthy. Besides, the determination of cost-effective yet high-performance sensors may be an adjusting act. Whereas high- precision sensors offer superior precision, they are regularly costly, making reasonableness an imperative in large-scale execution. Another issue is the integration of diverse equipment components, guaranteeing they work consistently without clashes in communication conventions.

To address these challenges, encourage investigate into power-efficient components, superior remote network conventions, and vigorous sensor calibration methods is fundamental. The improvement of secluded sensor frameworks that permit simple overhauls and substitutions might moreover move forward long-term usefulness.

7.2 SOFTWARE AND MODEL CONSTRAINTS

The victory of the mechanized indoor plant observing framework intensely depends on its computer program and algorithmic effectiveness. One of the most constraints is the computational control of implanted frameworks, which limits the complexity of the calculations that can be conveyed. Whereas progressed machine learning models might improve decision-making, they require noteworthy computational assets that low-power microcontrollers or edge gadgets may not bolster.

Another challenge lies in information handling and clamor sifting. Crude sensor information frequently contains irregularities, requiring preprocessing strategies to channel clamor whereas holding fundamental data. In the event that not optimized, these forms can present additional latency. Additionally, guaranteeing that the computer program remains lightweight whereas still advertising vigorous functionalities is vital for keeping up proficiency.

Compatibility is another concern, as coordination the framework with diverse equipment components may present firmware jumbles or convention irregularities. Program must be adaptable sufficient to back different sensor sorts and setups. Besides, real-time checking applications request effective memory administration to anticipate framework over- burden.

Security is additionally an imperative angle, especially on the off chance that the framework interfaces to the web or stores information in cloud-based stages. Executing solid encryption and verification instruments will help defend client information from potential cyber

7.3 REAL-TIME PROCESSING LIMITATIONS

Real-time information procurement and preparing are essential to a viable plant observing framework. In any case, a few impediments ruin accomplishing low-latency operation. One of the essential issues is sensor reaction time, as a few sensors may have characteristic delays in giving exact readings. Deferred or conflicting readings can influence the decision- making handle, driving to erroneous watering plans or error of plant wellbeing conditions.

Moreover, arrange inactivity may be a major challenge, particularly in setups where information is transmitted wirelessly. On the off chance that numerous sensor hubs communicate at the same time, blockage can happen, driving to information parcel misfortunes or transmission delays. The framework must actualize proficient communication conventions to guarantee that basic cautions reach the client without critical delay.

Another challenge is the adaptability of the framework. As more sensors or functionalities are included, preparing requests increment, potentially overloading the microcontroller or computational unit. This may lead to execution debasement, requiring optimization methods such as edge computing or various leveled handling models.

To overcome these impediments, real-time information buffering strategies, prescient analytics, and low-latency communication conventions ought to be investigated. Advance changes in cloud-based preparing seem moreover offer assistance offload computational errands whereas keeping up real-time responsiveness.

7.4 ACCESSIBILITY AND USABILITY CONSIDERATIONS

Since the venture is planned for outwardly impeded people, openness plays a imperative part in guaranteeing ease of use. One of the most concerns is the viability of voice-based alarms and commands. The clarity of the sound yield, reaction time, and customization alternatives for diverse clients must be carefully considered. If the voice prompts are vague, as well quick, or as well complex, clients may battle to decipher cautions accurately.

Another openness concern is gadget interaction. Clients who are outwardly impaired require interfacing that don't depend on visual signals, making touchscreen-based intelligent unreasonable. Instep, the framework ought to bolster voice commands, material input, or braille-based interfacing for improved ease of use.

Besides, joining multi-language back is vital to cater to a diverse range of clients. Customization alternatives, such as altering the speed and tone of voice notices, can upgrade client involvement. Natural clamor contemplations ought to moreover be calculated in, as outside sounds may meddled with voice-based cautions.

Conducting ease of use testing with genuine clients is basic to refining the framework and recognizing zones for change. Future improvements may incorporate gesture-based controls or AI-driven individual colleagues to supply an indeed more instinctive client involvement.

7.5 FUTURE ENHANCEMENTS

Long haul advancement of the plant monitoring system may present a few upgrades to move forward productivity, availability, and insights. One of the foremost promising headways is AI-driven plant wellbeing investigation, which seem foresee plant illnesses based on sensor readings and natural conditions. By joining computer vision or profound learning models, the framework might distinguish visual side effects of plant stretch, giving a more comprehensive conclusion.

Another major enhancement would be way better control administration methodologies, such as utilizing solar panels or energy-efficient sensors to expand battery life. Right now, battery confinements posture challenges for long-term, continuous operation, making supportability a key region for upgrade.

Extending the system's network choices, such as coordination it with shrewd domestic biological systems, would permit clients to control plant checking through voice colleagues like Alexa or Google Collaborator. Furthermore, cloud-based analytics and prescient modeling might empower proactive cautions, making a difference clients take preventive activities some time recently plant wellbeing falls apart.

Encourage upgrades seem incorporate multi-device synchronization, permitting clients to get to their plant observing information from different gadgets, counting savvy shows and wearables. Moving forward the AI show for infection and natural product discovery with real-time overhauls might also refine plant wellbeing appraisals, guaranteeing more exact suggestions for clients.

Chapter 8

Conclusion and Future Work

8.1 SUMMARY OF ACHIEVEMENTS

The Affordable Automated Indoor Plant Monitoring System for Visually Impaired Users effectively marries various sensors, wireless capability, and a usable interface to aid users in plant care. The system provides real-time feedback on environmental variables, including soil moisture, temperature, and lighting levels, which ensures that the plants are getting the best of care.

One of the most significant contributions of the project is the integration of AI-based leaf disease and fruit detection models that improve plant health analysis by flagging possible defects at an initial stage. Through this innovation, users are in a position to act early and enhance crop yields. The accessibility of the system through voice prompts and real-time notifications makes the system very easy to use even for visually impaired users, and the experience is seamless.

Furthermore, the project proves the ability to leverage low-cost and energy-efficient hardware while being as accurate as needed in monitoring plants. The combination with mobile applications means users can remotely access and monitor plant status at their convenience. Automated decision-making is also supported by the system to minimize human intervention in plant maintenance.

Through the use of machine learning algorithms, cloud connectivity, and sensor fusion, the project provides an intelligent, scalable plant monitoring solution. The merger of affordability, accessibility, and high-end AI-based features contributes to it being a worthy contribution to smart agriculture and home gardening solutions.

8.2 POTENTIAL IMPROVEMENTS

Though successful, the system can still be improved. A major area of improvement would be to optimize AI-based fruit and disease detection models by including a greater volume of data and more advanced training methods. This might enhance precision and make the system work for many different species of plants as well as across diverse environmental factors.

Another upgrade is improving battery life and energy efficiency, especially for distant deployment. Using predictive analytics to issue pre-emptive alerts for plant disease, irrigation, and environmental shifts would enhance the efficiency of the system further.

In addition, adding language support for voice notifications and mobile apps would make it more user-friendly for more users. Enhancing real-time synchronization between cloud storage and local devices would make the data consistent and have quicker response times.

8.3 FUTURE SCOPE OF THE PROJECT

The project can be extended to enable commercial agriculture, smart farming, and greenhouse automation. With the incorporation of IoT-based cloud storage for historical data analysis, the system could monitor long-term trends in plant health and optimize farming plans.

More interaction with agricultural research centers and precision farming companies would further develop the system for mass applications. Moreover, expanding the AI model to identify pests, soil nutrient deficiencies, and climate-based crop forecasts would make it much more useful in actual farming practices.

8.4 REAL-WORLD APPLICATIONS AND EXPANSION POSSIBILITIES

The Affordable Automated Indoor Plant Monitoring System has extensive uses outside of personal gardening, with far-reaching implications for smart agriculture, urban agriculture, and environmental monitoring. Its AI-based strategy can transform indoor and commercial agriculture by giving real-time insights into the health of plants, optimizing the use of resources, and minimizing manual intervention.

In agricultural business, the integration with automated irrigation, nutrient supply, and pest control can improve efficiency and sustainability in yields. Leaf disease and fruit detection using AI allow large farms to track the health of the plants, catch infections early, and automate the sorting and grading process, lowering food waste and enhancing quality assurance.

For indoor and urban gardening, this system can accommodate hydroponic farms, rooftop gardens, and intelligent greenhouses, providing the best growing conditions in small spaces. It can also help research centers study plant growth patterns under controlled environments, advancing precision farming technology.

A major societal contribution is that it can enable visually impaired individuals by offering sound-based instructions for plant maintenance, encouraging accessibility in agriculture.

Further development could encompass environmental sensors to monitor air quality and soil makeup, allowing for applications in climate resilience, smart cities, and ecological preservation.

As AI and IoT technologies advance, this system would become a totally independent plant care assistant, which will help in sustainable agriculture, food security, and intelligent environmental management worldwide.

Appendices

Appendix 1: Hardware Components and Their Functions ESP32 Microcontroller

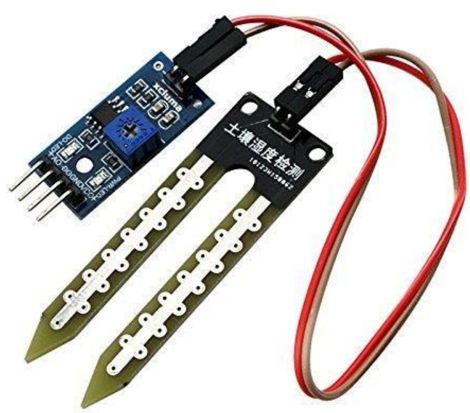
ESP32 is a low-power, low-cost microcontroller with onboard Wi-Fi and Bluetooth support, designed by Espressif Systems. It uses a dual-core Tensilica LX6 processor, which enables it to support several tasks at a time—a feature that is essential for real-time plant monitoring systems. It has a clock speed of up to 240 MHz and onboard 520 KB SRAM with external flash support. The board is generously provided with an extensive range of digital and analog input/output pins, touch sensors, ADCs, DACs, UART, SPI, I2C, and PWM interfaces, which render it extremely versatile for interfacing a variety of environmental sensors and actuators.

ESP32-CAM Module

ESP32-CAM is a small development board that integrates the ESP32 microcontroller and has an onboard OV2640 camera. It accommodates image resolution up to 1600x1200 and has a microSD card slot on the board for image saving. Even with its tiny form factor, it offers full wireless connectivity using Wi-Fi and Bluetooth, just like the standard ESP32 board. It also has GPIOs for connection to external devices and a flash LED for taking images in low light. The module can run at 3.3V and draws little power, which makes it appropriate for IoT and edge AI usage.

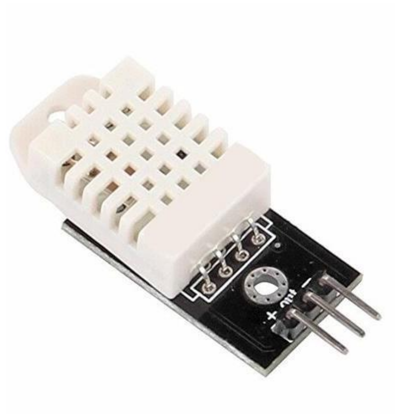
Soil Moisture Sensor

The Soil Moisture Sensor is a critical component in automated plant care systems, used to measure the volumetric water content in the soil. The sensor typically operates using a pair of exposed conductive probes that are inserted into the soil. It functions by measuring the resistance between the two probes: dry soil presents higher resistance, while wet soil offers lower resistance due to increased conductivity. In this project, an analog capacitive soil moisture sensor is used, which offers better resistance to corrosion compared to older resistive models. The sensor outputs analog values ranging from 0 to 4095, where lower values (typically below 1500) indicate dry soil and higher values (above 2500) suggest adequately moist conditions. It operates at a voltage of 3.3V to 5V, making it compatible with the ESP32 microcontroller.



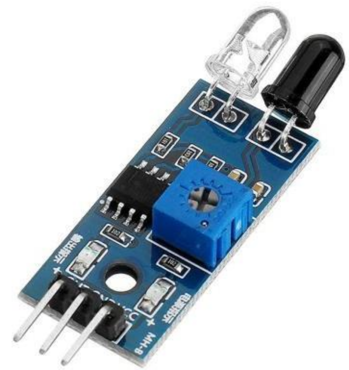
DHT22 Sensor (Temperature and Humidity Sensor)

The DHT22 is a digital temperature and humidity sensor known for its reliability and precision. It uses a capacitive humidity sensor and a thermistor to measure the surrounding air and outputs a digital signal on the data pin. The DHT22 can measure temperature in the range of -40°C to +80°C with an accuracy of ±0.5°C, and relative humidity from 0% to 100% with an accuracy of ±2-5%. It operates with a supply voltage of 3.3V to 6V, draws a very low current (~2.5 mA), and communicates over a single-wire serial interface, making it ideal for microcontroller-based systems like the ESP32. It also provides readings every 2 seconds (0.5 Hz), which is sufficient for plant monitoring applications.



Infrared (IR) Sensor

The Infrared (IR) sensor is a proximity sensor that detects objects based on infrared light reflection. It consists of an IR LED that emits light and a photodiode that detects reflected IR radiation. When an object comes into range, the IR light bounces back to the photodiode, causing a change in its output signal. Most IR sensors allow for adjustable detection range, typically from 2 cm to 30 cm, which can be fine-tuned using an onboard potentiometer. The sensor operates on a 3.3V to 5V power supply, making it compatible with the ESP32. It outputs a digital signal (HIGH or LOW) depending on whether an object is detected, which makes it easy to integrate into microcontroller-based systems.

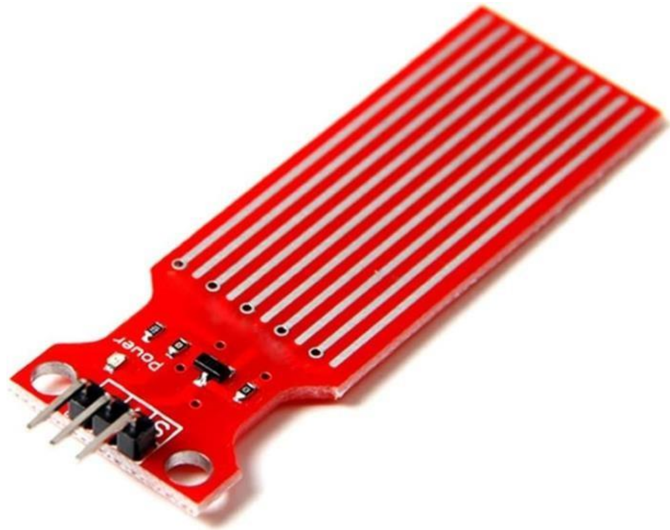


Light Dependent Resistor (LDR)

The Light Dependent Resistor (LDR), or photoresistor, is a light-dependent component whose resistance varies according to the amount of light striking it. Under bright light, the resistance of the LDR drops, enabling more current to flow through; in dim light, its resistance rises. This analog performance makes it perfectly suited for detecting changing light conditions. The LDR is normally set to a voltage of 3.3V to 5V and delivers analog readings that can be read through an ESP32's ADC (Analog-to-Digital Converter) pins. These readings are highly indicative of the surrounding lighting for the plant.

Water Level Sensor

Water Level Sensor is meant to detect the level of water in a reservoir or a container and plays a central role in the automated irrigation system. It normally works by employing a set of conductive or resistive strips to sense water height or capacitance changes. The sensor has analog output, typically between 0 and 4095, which is proportional to the water level. It runs on a supply of 3.3V to 5V and can be easily interfaced with the ESP32. These readings are subsequently utilized to ascertain whether there is enough water for irrigation or whether the reservoir must be refilled.



Appendix 2: Code Snippets

2.1 ESP32 Sensor Integration

```
// Read DHT Sensor
float temperature = dht.readTemperature();
float humidity = dht.readHumidity();
// Read Water Level Sensor
int waterLevel = analogRead(WATER_LEVEL_PIN);
// Read IR Sensors
int ir1 = digitalRead(IR_SENSOR_1);
int ir2 = digitalRead(IR_SENSOR_2);
// Read Soil Moisture Sensor
int soilMoisture = analogRead(SOIL_MOISTURE_PIN);
// Read LDR Sensor
int ldrValue = digitalRead(LDR_PIN);
```

2.2 Automated Watering Control

```
int soilMoisture = analogRead(SOIL_MOISTURE_PIN);
if (soilMoisture < 1800) {
  digitalWrite(RELAY_PIN, HIGH);  // Start watering
} else {
  digitalWrite(RELAY_PIN, LOW);   // Stop watering
}
```

2.3 ESP32 CAM Integration

Sending image captured from cam to flask server

```
String sendPhoto() {
  String getAll;
  String getBody;

  camera_fb_t * fb = NULL;
  fb = esp_camera_fb_get();
  if(!fb) {
    Serial.println("Camera capture failed");
    delay(1000);
    ESP.restart();
  }

  Serial.println("Connecting to server: " + serverName);

  if (client.connect(serverName.c_str(), serverPort)) {
    Serial.println("Connection successful!");
    String head = "--ESP32\r\nContent-Disposition: form-data; name=\"imageFile\"; filename=\"esp32-cam.jpg\"\r\nContent-Type: image/jpeg\r\n\r\n";
    String tail = "\r\n--ESP32--\r\n";

    uint16_t imageLen = fb->len;
    uint16_t extraLen = head.length() + tail.length();
    uint16_t totalLen = imageLen + extraLen;

    client.println("POST " + serverPath + " HTTP/1.1");
    client.println("Host: " + serverName);
    client.println("Content-Length: " + String(totalLen));
    client.println("Content-Type: multipart/form-data; boundary=ESP32");
    client.println();
    client.print(head);
```

```
uint8_t *fbBuf = fb->buf;
size_t fbLen = fb->len;
for (size_t n=0; n<fbLen; n=n+1024) {
  if (n+1024 < fbLen) {
    client.write(fbBuf, 1024);
    fbBuf += 1024;
  }
  else if (fbLen%1024>0) {
    size_t remainder = fbLen%1024;
    client.write(fbBuf, remainder);
  }
}
client.print(tail);

esp_camera_fb_return(fb);

int timoutTimer = 10000;
long startTimer = millis();
boolean state = false;

while ((startTimer + timoutTimer) > millis()) {
  Serial.print(".");
  delay(100);
  while (client.available()) {
    char c = client.read();
    if (c == '\n') {
      if (getAll.length()==0) { state=true; }
      getAll = "";
    }
    else if (c != '\r') { getAll += String(c); }
    if (state==true) { getBody += String(c); }
    startTimer = millis();
  }
```

```
    if (getBody.length()>0) { break; }
  }
  Serial.println();
  client.stop();
  Serial.println(getBody);
}
else {
  getBody = "Connection to " + serverName +  " failed.";
  Serial.println(getBody);
}
return getBody;
}
```

2.4 Retrieve image using Flask

```
# Flask endpoint
@app.route('/upload', methods=['POST']) def
predict_from_camera():
    if 'imageFile' not in request.files:
        return jsonify({"error": "No image file provided"}), 400
    file = request.files['imageFile']
    nparr = np.frombuffer(file.read(), np.uint8)
    img = cv2.imdecode(nparr, cv2.IMREAD_COLOR) if img is
    None:
        return jsonify({"error": "Failed to decode image"}), 422
    results = process_image_for_prediction(img) return
    jsonify(results), 200
```

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